

Temporal Separation with Entropy Regularization for Knowledge Distillation in Spiking Neural Networks

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Abstract

Spiking Neural Networks (SNNs), inspired by the human brain, offer significant computational efficiency through discrete spike-based information transfer. Despite their potential to reduce inference energy consumption, a performance gap persists between SNNs and Artificial Neural Networks (ANNs), primarily due to current training methods and inherent model limitations. While recent research has aimed to enhance SNN learning by employing knowledge distillation (KD) from ANN teacher networks, traditional distillation techniques often overlook the distinctive spatiotemporal properties of SNNs, thus failing to fully leverage their advantages. To overcome these challenges, we propose a novel logit distillation method characterized by temporal separation and entropy regularization. This approach improves existing SNN distillation techniques by performing distillation learning on logits across different time steps, rather than merely on aggregated output features. Furthermore, the integration of entropy regularization stabilizes model optimization and further boosts the performance. Extensive experimental results indicate that our method surpasses prior SNN distillation strategies, whether based on logit distillation, feature distillation, or a combination of both. Our project is available at <https://github.com/yukairong/TSER>.

1. Introduction

Inspired by the neural firing mechanisms observed in biological systems, Spiking Neural Networks (SNNs) [37] are considered a promising alternative to traditional Artificial Neural Networks (ANNs) due to their superior energy efficiency. Unlike ANNs, which use continuous activation val-

ues for information transmission[31], SNNs transmit and process information through discrete spike events [6, 38], with neurons generating spikes only when their membrane potential exceeds a threshold. This binary, event-driven approach allows SNNs to run efficiently on neuromorphic hardware [6, 42], accumulating synaptic inputs effectively and avoiding unnecessary computations related to zero input or activation [9, 14]. Given their event-driven dynamics and the biomimetic properties of spatiotemporal neuron activity [44, 45], SNNs demonstrate remarkable energy efficiency, robust adaptive learning capabilities [13, 41, 58], and ultra-low power consumption, showing significant potential for computational intelligence applications tasks [44]. Despite the inherent advantages of SNNs, their performance in common tasks such as image classification [27], object segmentation [43], and natural language processing [23] still lags behind that of ANNs. This gap largely stems from the limitations of current training methods and structural constraints in SNNs. In contrast to ANNs, SNNs cannot directly utilize backpropagation (BP) for deep network training. Moreover, attempts to directly adapt ANN methodologies to SNNs encounter compatibility issues, preventing SNNs from fully realizing their theoretical advantages.

Currently, conversion-based and learning-based approaches are two common methods for training SNNs. The former approaches seek to utilize knowledge from ANNs by transferring the parameters of a pre-trained ANN to an equivalent SNN [1, 10, 33, 34]. These methods often necessitate a significant number of time steps to achieve accuracy levels comparable to those of the original ANN [2, 46]. If the number of inference time steps is reduced, the model's capacity for effective information transfer may diminish, resulting in a decline in overall performance. To improve the performance of SNNs and minimize inference time steps, learning-based approaches have been employed

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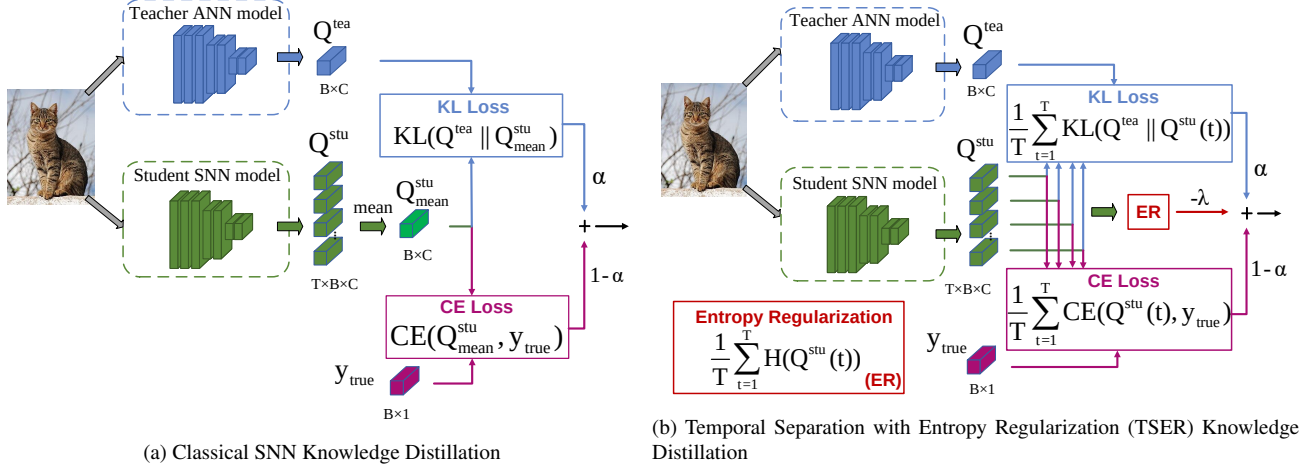


Figure 1. Illustration of the Classical SNN KD and Our Proposed Method. In our approach, we remove the temporal dimension fusion operation present in classical SNN KD and apply a temporal separation strategy to focus the learning process on outputs at individual time steps. The original calculations of KL Loss and CE Loss are adapted to compute the loss for each time step’s output, followed by averaging. Additionally, we incorporate an entropy regularization term to guide the learning direction away from erroneous knowledge. Here, T , B , and C represent the time steps, batch size, and number of classes, respectively.

to facilitate training during inference. These approaches leverage Backpropagation Through Time (BPTT) for effective training [48, 57]. Motivated by the work [25], several studies have utilized surrogate gradient estimation as a method to address the non-differentiability inherent in SNNs. This foundational research has enabled the direct training of CNNs and transformer-based SNNs, achieving strong performance on static datasets with reduced inference time steps. To further reduce the performance gap with ANNs, recent work has explored the inclusion of modules such as attention mechanisms to enhance SNN capabilities [28, 40, 52, 53]. While effective, these additional modules increase computational consumption, counteracting the inherent low-energy benefits of SNNs. Knowledge Distillation (KD) provides a potential solution, where SNNs are guided by a more capable ANN teacher model, addressing the inference burden associated with these modules. Prior works such as KDSNN [50] and LaSNN [24] employ hierarchical knowledge distillation [5] to facilitate the training of SNNs by leveraging the rich semantic information from the ANN teacher model. Recent approaches like BKD-SNN [51] move beyond traditional layer-wise distillation by integrating blurred knowledge distillation techniques with logit distillation, demonstrating significant improvements in performance across both CNNs and Transformer-based models.

In preliminary experiments, we observed that current SNN distillation architectures are not optimally designed for the unique structure of SNNs. As depicted in Fig. 1a, conventional SNN KD techniques distill the mean of outputs across time steps. However, the inherent nature

of SNN processing implies that significant information is embedded within the temporal dimension. Employing a mean approach for fusion prediction during classification resembles a voting selection process. Therefore, distributing the distillation of outputs across various time steps during the training phase can substantially alleviate the limitations that currently hinder distillation performance. To mitigate the impact of outliers on network performance and to rectify overly confident incorrect semantics produced by the teacher network, we introduce an entropy regularization term as a constraint. We conduct extensive experimental analyses to assess its efficacy. Furthermore, we evaluate our proposed method on several datasets, including CIFAR10 [26], CIFAR100 [26], and ImageNet [8]. The results of our experiments demonstrate that this method achieves state-of-the-art (SOTA) performance. In summary, our contributions are threefold:

- We introduce a temporal separation strategy into the SNN distillation method, effectively addressing the incompatibility issues present in previous SNN distillation frameworks and yielding a more effective universal distillation architecture.
- We integrate entropy regularization to correct the overly confident errors from the teacher network, thereby maximizing the performance of the logit-based distillation method.
- We conduct comprehensive evaluations across multiple datasets, showing that our proposed approach outperforms previous state-of-the-art (SOTA) methods.

2. Related Work

Conversion-based Methods. To fully leverage the knowledge acquired from pre-trained ANNs, conversion methods directly transfer the learned parameters from an ANN to the corresponding SNN, avoiding the difficulties of training SNNs from scratch. Several studies [19, 20] replace the activation functions in ANNs with spiking neurons and introduce optimizations like weight normalization [12] to generate corresponding SNNs. The core idea of these methods is to closely align the outputs of the ANN and SNN, maximizing the utilization of the pre-trained ANN knowledge. While later works [5, 10, 12] achieve nearly lossless accuracy conversion, they often involve long inference times and fail to retain the spatiotemporal characteristics inherent to spike-based processing, significantly limiting the practical applicability of converted SNNs.

Learning-based Methods. The development of surrogate gradient training methods [56] has resolved the challenge of training non-differentiable SNNs directly. Initially, [39] used a surrogate gradient approach to construct deep SNN models, achieving competitive accuracy on neuromorphic datasets. The DCT-SNN [16] applied frequency domain techniques to reduce the number of inference time steps. To support deeper SNN training, [15] introduced SEW ResNet to address gradient vanishing and exploding issues in Spiking ResNet [60], extending SNNs to architectures exceeding 100 layers. TA-SNN [52] introduced a temporal attention mechanism, paving the way for future attention mechanism advancements in the field. MA-SNN [53] went beyond single-dimensional information enhancement by proposing a temporal-channel-spatial attention mechanism. In Transformer architectures, models like SpikeFormer [61], SpikingFormer [62], and Meta-SpikerFormer [63] demonstrated outstanding classification performance through the design and improvement of innovative spike-based self-attention blocks.

Knowledge Distillation for SNN. Knowledge Distillation (KD), a well-established transfer learning technique, has proven effective across various tasks [3, 4, 22]. Recent studies [17, 30, 32, 47, 59] have applied KD to SNN training, using either large isomorphic SNNs to extract knowledge for smaller SNNs or leveraging the logits from pre-trained ANNs to train SNNs. KDSNN [50] utilizes a joint distillation approach based on logits and features, showing effectiveness across multiple datasets. Xu *et al.* [49] explore biologically inspired structure learning using reverse knowledge distillation. Furthermore, LaSNN [24] proposes a hierarchical feature distillation framework that achieves accuracy comparable to ANNs on the Tiny ImageNet dataset. BKDSNN [51] improves SNN performance

on complex datasets by using blurred knowledge to replicate ANN features. Despite the advantages of these distillation methods, as shown by the experiments in Fig. 2, current SNN distillation frameworks have yet to fully account for the unique characteristics of SNNs. Developing tailored distillation paradigms could enhance model accuracy in a simple and effective way.

3. Methodology

In this section, we start from preliminaries, introducing the fundamental concepts of SNNs and SNN distillation. Then we introduce our knowledge distillation approach that incorporates temporal separation with entropy regularization.

3.1. Preliminaries

Spiking Neuron Model. In SNNs, the fundamental computational unit is the spiking neuron, which serves as an abstract model of the dynamics of biological neurons. Currently, the Leaky Integrate-and-Fire (LIF) model is one of the most commonly used spiking neuron models, as it strikes a balance between simplified mathematical formulation and the complex dynamics of biological neurons. Therefore, in this study, we employ the LIF model as the foundational neuron model for the student SNN. Mathematically, a LIF neuron can be represented by Eq. 1.

$$\mu \frac{du(t)}{dt} = -(u(t) - u_{\text{reset}}) + I(t). \quad (1)$$

Where μ denotes the time constant of the membrane potential, $u(t)$ represents the membrane potential at time t , u_{reset} is the resting potential of the neuron, and $I(t)$ is the presynaptic input at time t . Based on this differential equation, the discrete-time and iterative mathematical representation of the LIF-SNN can be described as follows:

$$\begin{aligned} V(t) &= H(t-1) + \frac{1}{\mu}[I(t-1) - (H(t-1) - u_{\text{reset}})] \\ S(t) &= \Theta(V(t) - v_{th}) \\ H(t) &= u_{\text{reset}} \cdot S(t) + V(t) \cdot (1 - S(t)) \end{aligned} \quad (2)$$

The Heaviside step function Θ is defined as:

$$\Theta(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0. \end{cases} \quad (3)$$

Among these, $H(t-1)$ denotes the membrane potential following spike generation at the previous time step, while $I(t)$ and $V(t)$ represent the input and the updated membrane potential at time step t , respectively. Furthermore, v_{th} is the threshold that determines whether $V(t)$ should fire a spike or remain silent, and $S(t)$ indicates the spike sequence generated after triggering the action potential at time step t .